

Thanh Vuong Dang

Viettel High Tech

Ho Chi Minh, Viet Nam

Email: dangthanhvuong97@gmail.com

ORCID: 0009-0003-3039-1428

Personal page: vuongdang97.github.io

LinkedIn: <https://www.linkedin.com/in/vuong-dang-thanh-278194181/>

Research Interests

Numerical Analysis, High Performance Computing, Probability and Statistics.

Education

Ph.D. in Applied Mathematics 2020–2023

University of Paris 13

Thesis: **Parallel in time algorithms for internal wave controlled problem**

Link thesis: <https://hal.science/tel-04701863>

Supervisor: Prof. Laurence Halpern, MCF. Bérangère Delourme

M.Sc. in Applied Mathematics 2019–2020

University of Orleans

Thesis: **Schwarz waveform relaxation algorithms for optimal control problem constrained by the wave equation**

GPA: 17/20

Supervisor: Prof. Laurence Halpern, MCF. Bérangère Delourme

B.Sc. in Mathematics 2015–2019

University of Sciences, Ho Chi Minh, Viet Nam

Thesis: **Morrey Spaces**

GPA: 9.16/10

Supervisor: Prof. Duong Minh Duc

Research Experience

Algorithm Researcher - Viettel High Tech 04/2026–Present

- Developing algorithm to optimize the 5G advanced and 6G.
- Developing information geometry to find potential statistical estimator for ISAC (Integrated Sensing and Communication).
- Collaborated with FPGA engineering team.

Principal Researcher - Simidea R&D 03/2025–06/2025

- Develop Jpythmen library (Java) for Johnson–Champoux–Allard model and Biot model in impedance tube.
- Simulate the physical experiment using Groovy console, Java, Symplegma Development Environment (SDE) and COMSOL to compute absorption, reflexion, transmission and dissipation coefficient of solid materials.

- Implement inverse characterization to identify Lamé coefficients and density of solid materials based on real experiment data conducted at company utilizing Differential Evolutionary Algorithm.
- Link: <https://simidea.es/>

Academic research position - University Paris 13, France

04/2019– 12/2023

- Solve optimization problems with internal wave constraints by Finite Volume Method with MATLAB, Python and C/C++ on Ubuntu.
- Solve optimal control problems of wave equation with boundary control data by Finite Difference Method with MATLAB, Python and C/C++ on Ubuntu.

Publications

1. Thanh Vuong Dang, with L.Halpern and B.Delourme Time Parallelization of a Distributed Optimal Control Problem for the Wave Equation,
28th International Conference on Domain Decomposition Methods at KAUST, Saudi Arabia.
Link: https://www.ddm.org/DD28/proceedings/DD28_271.pdf.
2. Thanh Vuong Dang, On the boundary controlled wave equation,
Link: <https://hal.science/hal-04463449>,

Awards and Honors

- National scholarships to develop mathematics from 2016-2019
Vietnam Institute for Advanced Study in Mathematics (VIASM).

Teaching Experience

Teaching Assistant – University of Sciences, Ho Chi Minh city

2019–2020

- Analysis 1.
- Graded 40 students' assignments.

Academic Mentor – Funix, Learn with mentor

2026–Present

- General Mathematics including graph theory, combinatorics, analysis, probability, statistics, algebra.
- Physics including classical dynamics and fluid dynamics.
- Computer sciences including introduction to programming, data structures and Object Oriented Programming.

Invited Talks

- 06/2023: An internal controlled wave problem: Schwarz and Dirichlet-Neumann parallel in time methods - Workshop on Assimilation, Control and Computational Speedup, LAGA.
- 09/2022: Schwarz method in time for internal controlled wave problem - Research school: Domain Decomposition Method for Optimal control problem.

- 02/2022: An introduction to optimal control and controlled wave problem - Doctoral Seminar, LAGA.
- 09/2021: Parallelisation of an optimal control problem with wave constraints - Parallelization and Control day, LAGA.

Technical Skills

Programming: C/C++, Java, Python, MATLAB, Groovy

Software: LaTeX, Maple, COMSOL

Database: SQL, Pandas

Languages

English: C1

France: B1

References

Prof: Laurence Halpern

Professor Emeritus / Paris 13 University

Email : halpern@math.univ-paris13.fr

Prof: Luis Sanchez Ricart

Professor / University of Cartagena

Founder at Simidea R&D

Email : l.sanchez@simidea.es

MCF: Bérangère Delourme

MCF/ Paris 13 University

Email : delourme@math.univ-paris13.fr